Bonded Gold Sand

Bonded metallic polyester, TGIC-free option · RAL 1036

RAL 1036 · Pearl gold

Product	Code • series	Colour reference	Document
Bonded Gold Sand 邦定金砂	BL-M102 BL-M Bonded Metallics	RAL 1036 · Pearl gold	TDS BL-M102 Rev. 2026-10

P1000 · tested to AAMA 2603 / QUALICOAT I test items on request — no product certificate held

1. Description

A bonded metallic for aluminium profiles, façade panels and appliance trims where an anodised-gold look is wanted without running an anodising line. The metallic pigment is bonded to the base particle, so the pigment-to-base ratio stays constant through transport, spraying and reclaim. It is not a Class-A automotive face finish: the effect reads as depth, not as a mirror.

2. Typical properties

Property	Value	Method
Chemistry	Bonded metallic polyester, TGIC-free option	—
Colour reference	RAL 1036 Pearl gold — approval on a sprayed panel against a physical chip	RAL
Finish and gloss	Sand, 8–15 @60°	ASTM D523
Film build	70–90 µm	ISO 2808
Density	1.55–1.65 g/cm³	ISO 8130-2/3
Coverage	7.8 m²/kg at 100 % film · 7.4 m²/kg planning value at 95 % utilisation (with reclaim)	calculated
Cure schedule	200 °C × 10 min (object temperature)	object temp., not oven set point
Particle size D50	30–45 µm	ISO 8130-1 / -13
Stoving loss	≤ 1–2 %	ISO 8130-7
Storage stability	1 month at 35–40 °C	ISO 8130-8

3. Performance data

Test	Method	Typical result
1. Adhesion, cross-cut	ISO 2409 / ASTM D3359	0–1 (100/100 on aluminium with a conversion coating)
2. Impact, direct and reverse	ASTM D2794	50 kg·cm, no cracking
3. Pencil hardness	ASTM D3363	H (2H typical)
4. Gloss 60°	ASTM D523	Sand texture, 8–15
5. Colour difference	ASTM D2244	ΔE ≤ 1.5 same batch
6. Neutral salt spray	ASTM B117 / ISO 9227	500 h; 1,000 h on request with conversion coating at 80 µm
7. QUV-A	ASTM G154	1,000 h, gloss retention ≥ 50 %
8. Chemical resistance	ASTM D1308	Per application — tell us the exposure

Measured to the method listed, on production batches or the third-party report held for this system. Not a certificate: acceptance values are fixed in the trial report and the purchase specification.

4. Application and cure

01 Aluminium pre-treatment

Degrease → alkaline or acid etch → rinse → chromium-free conversion coating → DI rinse → dry. On bare aluminium without a conversion coating, plan on roughly half the salt-spray hours.

02 Steel pre-treatment

Degrease → zinc or iron phosphate → rinse → DI rinse → dry. Blast to Sa 2½ for heavy corrosion duty.

03 Application

Corona, 60–90 kV, gun-to-part 150–250 mm, 70–90 µm in one pass. Bonded metallics want a steady reciprocator: hand-spraying the same part twice gives two different metallic orientations.

04 Cure

Object temperature 200 °C for 10 min. Measure the part, not the oven display — a 20 °C overshoot darkens a metallic shade.

05 Reclaim

Overspray can be reclaimed and blended back: the bonded pigment does not separate, so the colour does not shift. Total utilisation above 95 %.

5. Packaging, storage and ordering

1. Standard pack: 20 kg net, polyethylene-lined carton. Other pack sizes on request.
2. Store below 30 °C, dry, out of direct sunlight, in the unopened pack. Keep away from ovens and hot lines.
3. Shelf life: 12 months from the production date in the original pack. Storage stability is verified at 35–40 °C for one month (ISO 8130-8).
4. A batch test report accompanies every delivery; a retained sample is kept for 12 months.
5. Sample in 3 days; production lead time confirmed per colour and pack size at order.
6. Colour match ΔE ≤ 1.5 within a batch. Cross-batch tolerance is agreed in the purchase specification.
7. Ask for: SDS · third-party test report · batch test report · physical colour chip.

6. Health and safety

1. Not classified as dangerous goods for transport. The SDS is available for every product — ask for it with the TDS.
2. The powder is a combustible dust. Earth the workpiece, the gun and the booth; no open flame or hot work in the booth; extract booth air; avoid dust clouds when emptying or cleaning.
3. Wear a P2 particulate mask, goggles and gloves when handling dry powder or cleaning the booth.
4. Cure volatiles: extract the oven and let parts cool before handling. Do not enter the oven zone without ventilation.

7. Also in this range

Code	Product	Colour	Film build	Cure
BL-M103	Bonded Mirror Silver	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min

8. Document control

Sheet	Revision	Issued	Supersedes	Languages
BL-M102 TDS	2026-10	2026-10	None — first issue	English + 中文 (4 pp.)

9. Limits of this document

What this product is not for:

— Not for a 10-year exterior colour match: the reference class is P1000. Use BL-A202 for curtain wall and coastal work.

— Not a mirror. For a chrome-like face, BL-M103 plus its clear coat is the right product.

— Touch-up on an assembled part reads as a patch under direct light. Plan the touch-up into the line, not after installation.

1. Values in this sheet are typical for the product and were measured to the methods listed. They are not a certificate and not a warranty of fitness for a particular purpose. Acceptance values are fixed in the trial report and the purchase specification.
2. Colour: RAL references only. RAL does not define sRGB values, so the swatch in this sheet is a digital approximation — approval is made on a sprayed panel against a physical RAL chip.
3. No AAMA 2603 / 2604 / 2605, QUALICOAT, GSB or UL product certificate is held. We test to those methods and can supply third-party reports and in-house batch reports; if a tender requires the certificate itself, we will say plainly that we do not have it.
4. The 4,000 h salt spray and 10-year South Florida results shown elsewhere on our site belong to the fluorocarbon (PVDF) system, not to this powder product.

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Ask for: SDS · third-party test report · batch report · physical colour chip · sample in 3 days

邦定金砂

邦定金属聚酯，可选 TGIC-free · RAL 1036

RAL 1036 · 珍珠金

产品	编号 • 系列	参照颜色	文件
Bonded Gold Sand 邦定金砂	BL-M102 BL-M 邦定金属	RAL 1036 · 珍珠金	TDS BL-M102 Rev. 2026-10

P1000 · 可按 AAMA 2603 / QUALICOAT I 测试项目送检（可索取报告）——不持有产品认证证书

1. 产品描述

用于铝型材、幕墙板与家电饰件，想要“氧化金”观感但不想走氧化线。金属颜料键合在基粉颗粒上，运输、喷涂、回收全程比例不变。它不是汽车 Class-A 外观面漆：效果是深度感，不是镜面。

2. 典型性能

项目	数值	方法
体系	邦定金属聚酯，可选 TGIC-free	—
参照颜色	RAL 1036 珍珠金——以喷板对照实物色卡确认	RAL
表面与光泽	砂纹 8–15（60°）	ASTM D523
膜厚	70–90 µm	ISO 2808
比重	1.55–1.65 g/cm³	ISO 8130-2/3
覆盖率	100% 成膜 7.8 m²/kg · 按 95% 利用率（含回收）规划值 7.4 m²/kg	计算值
固化条件	200 °C × 10 min（工件温度）	测工件温度，非炉膛设定
粒径 D50	30–45 µm	ISO 8130-1 / -13
烘烤失重	≤ 1–2 %	ISO 8130-7
贮存稳定性	35–40 °C 下 1 个月	ISO 8130-8

3. 性能数据

检测项目	方法	典型结果
1. 划格附着力	ISO 2409 / ASTM D3359	0–1（铝件经转化膜处理可达 100/100）
2. 正反冲击	ASTM D2794	50 kg·cm 不裂
3. 铅笔硬度	ASTM D3363	H（典型 2H）
4. 光泽 60°	ASTM D523	砂纹 8–15
5. 色差	ASTM D2244	同批 ΔE ≤ 1.5
6. 中性盐雾	ASTM B117 / ISO 9227	500 h；经转化膜、膜厚 80 µm 可订 1000 h
7. QUV-A 加速老化	ASTM G154	1000 h，保光率 ≥ 50%
8. 耐化学性	ASTM D1308	按工况——请告知接触介质

按所列方法在量产批次或本体系所持第三方报告上测定。不是证书：验收值以试粉报告与采购规格为准。

4. 施工与固化

01　铝件前处理

脱脂 → 碱蚀或酸蚀 → 水洗 → 无铬转化膜 → 纯水洗 → 干燥。裸铝不做转化膜时，盐雾时数按大约一半估算。

02　钢件前处理

脱脂 → 锌系或铁系磷化 → 水洗 → 纯水洗 → 干燥。重防腐工况喷砂至 Sa 2½。

03　喷涂

电晕 60–90 kV，枪距 150–250 mm，一次上膜 70–90 µm。邦定金属要求往复机速度稳定：同一工件手喷两遍，金属排列方向会不一致。

04　固化

工件温度 200 °C × 10 min。测工件温度，不测炉膛显示温度——超温 20 °C 金属色就会变深。

05 回收

过喷粉可回收掺用：邦定颜料不分离，色相不漂移。总利用率 95% 以上。

5. 包装、贮存与订货

1. 标准包装：20 kg 净重，纸箱内衬聚乙烯袋。其他包装规格可订。
2. 30 °C 以下、干燥、避免阳光直射，保持原包装未开封。远离烘道与高温工位。
3. 保质期：原包装自生产日起 12 个月。贮存稳定性按 ISO 8130-8 于 35–40 °C 验证一个月。
4. 每批附批次检测报告；留样保存 12 个月。
5. 3 天出样；生产交期按颜色与包装规格在订单时确认。
6. 同批色差 ΔE ≤ 1.5；跨批公差在采购规格中约定。
7. 可索取：SDS · 第三方检测报告 · 批次检测报告 · 实物色卡。

6. 健康与安全

1. 不属于运输危险品。每个产品均提供 SDS（安全数据表），可与 TDS 一并索取。
2. 粉末为可燃性粉尘。工件、喷枪与喷房必须接地；喷房内禁止明火与动火作业；喷房须抽风；倒粉与清房时避免扬尘。
3. 处理干粉或清理喷房时，佩戴 P2 防尘口罩、护目镜与手套。
4. 固化挥发物：烘道须抽风；工件冷却后再搬运。无通风不得进入烘道区域。

7. 同系列其他产品

编号	产品	颜色	膜厚	固化
BL-M103	邦定镜面银	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min

8. 文件控制

文件	版本	签发	替代版本	语言
BL-M102 TDS	2026-10	2026-10	无——首版	英文 + 中文（共 4 页）

9. 本文件的边界

这款产品不适合用在：

— 不适用于 10 年户外保色要求：参照等级为 P1000。幕墙与沿海项目请用 BL-A202。

— 不是镜面。要电镀铬观感，请用 BL-M103 配套透明罩光。

— 已组装工件上补喷，在直射光下就是一块补丁。补喷要安排在产线上，不要留到安装之后。

1. 本表数值为该产品的典型值，按所列方法测定。它们不是证书，也不构成对特定用途适用性的保证。验收值以试粉报告与采购规格为准。
2. 颜色：仅作 RAL 参照。RAL 未定义 sRGB 数值，本表色块为数字近似——以喷涂样板对照实物 RAL 色卡确认。
3. 我们不持有 AAMA 2603 / 2604 / 2605、QUALICOAT、GSB 或 UL 产品认证证书。我们按这些方法检测，可提供第三方报告与厂内批次报告；若招标硬性要求证书本身，我们会直接说明没有。
4. 我们网站上 4000 小时盐雾与 10 年南佛罗里达曝晒的结果属于氟碳（PVDF）体系，不属于本粉末产品。

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可索取：SDS · 第三方检测报告 · 批次报告 · 实物色卡 · 3 天出样